INTERNATIONAL SEARCH REPORT

International application No.
PCT/IB2019/060432

A. CLASSIFICATION OF SUBJECT MATTER

A61M 16/06 (2006.01) A61M 16/08 (2006.01)

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

PATENW and A61M16/0683, A61M16/0833, A61M16/0875, A61M16/-, A61M16/0666/low, A61M39/10/low, F16L11/- and keywords: tube, stiff, textile, foam, multiple, length, circumference, part, patient, side, longitudinal, seam, reinforce, connect, wire, crown, top, head, port, internal, prevent, occlusion, contact, wall, protrusion and similar terms.

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
	Documents are listed in the continuation of Box C	

[X] Further documents are listed in the continuation of Box C [X] See patent family annex

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"D" document cited by the applicant in the international application

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
27 February 2020	27 February 2020

Name and mailing address of the ISA/AU

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Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☐ Claims Nos.:

because they relate to subject matter not required to be searched by this Authority, namely:

the subject matter listed in Rule 39 on which, under Article 17(2)(a)(i), an international search is not required to be carried out, including

2. ☐ Claims Nos.:

because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:

3. ☐ Claims Nos:

because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a)

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:

See Supplemental Box for Details

1. ☒ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.
2. ☐ As all searchable claims could be searched without effort justifying additional fees, this Authority did not invite payment of additional fees.
3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

4. ☐ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims; it is covered by claims Nos.:

Remark on Protest

☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.

☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.

☒ No protest accompanied the payment of additional search fees.

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	WO 2017/124155 A1 (RESMED LIMITED) 27 July 2017 paragraphs 152, 155, 160, 221, 296, 300 and 473. Figure 3C	1 - 22, 25, 26, 78, 95
X Y	US 2015/0083136 A1 (KONINKLIJKE PHILIPS N.V.) 26 March 2015 figures 1, 5 and 10, paragraphs 33, 34, 44, 45, 47 figures 1, 5 and 10, paragraphs 33, 34, 44, 45, 47	1 - 10, 12 - 14, 25 - 28, 30, 33, 38 - 52, 54 - 60, 78, 95 43, 44
A	US 2012/0325219 A1 (SMITH) 27 December 2012	1
A	US 2011/0247619 A1 (FORMICA et al.) 13 October 2011 figure 13, paragraph 74	1
X	WO 2008/011682 A1 (RESMED LIMITED) 31 January 2008 figures 1-1 and 3-6a, page 17 lines 3 to 15	27 - 39, 45 - 47, 95
X Y	US 2017/0333662 A1 (RESMED LIMITED) 23 November 2017 figures 6, 9, 10, paragraph 188 figures 6, 9, 10, paragraph 188	27 - 41, 45 - 47, 95 43, 44
X Y	US 2012/0152255 A1 (BARLOW et al.) 21 June 2012 figures 3-1 and 3-2-1 to 3-2-3 figures 3-1 and 3-2-1 to 3-2-3	27 - 29, 32 - 39, 45 - 47, 95 43, 44
A	US 2009/0211579 A1 (SMITH et al.) 27 August 2009 figure 1a	27
A	WO 2018/029638 A1 (FISHER & PAYKEL HEALTHCARE LIMITED) 15 February 2018 figures 8C and 8D	27
A	US 2017/0082223 A1 (GARRETT et al.) 23 March 2017 figure 4	48
P,A	WO 2019/122133 A1 (INTERSURGICAL AG) 27 June 2019	48
A	US 2452047 A (HAMBLIN) 26 October 1948	48
X	US 2016/0008564 A1 (KONINKLIJKE PHILIPS N.V.) 14 January 2016 figure 6, paragraph 46, figure 13 paragraph 70	61 - 77, 79 - 94
A	US 2016/0317776 A1 (RESPCARE, INC.) 03 November 2016 figure 4, paragraphs 61 and 64	61
A	US 2008/0011305 A1 (CHANDRAN et al.) 17 January 2008 figure 5	61
P,A	WO 2019/185474 A1 (KONINKLIJKE PHILIPS N.V.) 03 October 2019 figures 7A to 7D	61

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	US 6135108 A (HOENIG) 24 October 2000 figure 4, column 4 lines 41 to 64	79
A	US 2008/0097350 A1 (BELL et al.) 24 April 2008 figures 8A to 9B	79
A	US 2012/0037162 A1 (BOUSSIGNAC) 16 February 2012 paragraph 75, figure 2	79

Supplemental Box

Continuation of: **Box III**

This International Application does not comply with the requirements of unity of invention because it does not relate to one invention or to a group of inventions so linked as to form a single general inventive concept.

This Authority has found that there are different inventions based on the following features that separate the claims into distinct groups:

- Claims 1 to 26, 78 and 95 (in part) directed to a positioning and stabilizing structure. The feature of one of the inferior tube portion and superior tube portion having a greater rigidity than the other of the inferior tube portion and superior tube portion is specific to this group of claims.

- Claims 27 to 47 and 95 (in part) directed to a positioning and stabilizing structure. The feature of first, second and third elongate lengths of material joined along their edges is specific to this group of claims.

- Claims 48 to 60 directed to a positioning and stabilizing structure. The feature of the seam comprises rigidising elements embedded within or layered on the seam to provide rigidity to the gas delivery tube is specific to this group of claims.

- Claims 61 to 77 directed to a positioning and stabilizing structure. The feature of the crown piece connector comprising a material that is more rigid than the flexible material of the crown connector is specific to this group of claims.

- Claims 79 to 94 directed to a positioning and stabilizing structure. The feature of a crown connector including an occlusion-preventer and a ramp sloping downwardly towards the occlusion-preventer is specific to this group of claims.

PCT Rule 13.2, first sentence, states that unity of invention is only fulfilled when there is a technical relationship among the claimed inventions involving one or more of the same or corresponding special technical features. PCT Rule 13.2, second sentence, defines a special technical feature as a feature which makes a contribution over the prior art.

When there is no special technical feature common to all the claimed inventions there is no unity of invention.

In the above groups of claims, the identified features may have the potential to make a contribution over the prior art but are not common to all the claimed inventions and therefore cannot provide the required technical relationship. The only feature common to all of the claimed inventions and which provides a technical relationship among them is gas delivery tube being constructed and arranged to contact, in use, at least a region of the patient's head superior to an otobasion superior of the patient's head

However this feature does not make a contribution over the prior art because it is disclosed in:

WO 2017/124152 A1 (RESMED LIMITED) 27 July 2017 (see Figure 3B for example; Figure 3B also shows a crown connector 3600, which is an additional common feature between independent claims 61 and 79).

Therefore in the light of this document this common feature cannot be a special technical feature. Therefore there is no special technical feature common to all the claimed inventions and the requirements for unity of invention are consequently not satisfied *a posteriori*.

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This Annex lists known patent family members relating to the patent documents cited in the above-mentioned international search report. The Australian Patent Office is in no way liable for these particulars which are merely given for the purpose of information.

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